

Unit 14, Lesson 2: Determining Sample Space Using Outcome Tables

Benchmark

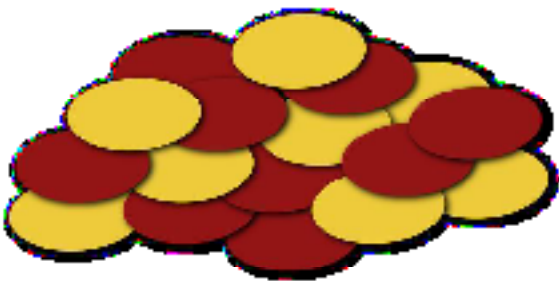
MA.8.DP.2.1 Determine the sample space for a repeated experiment.

Learning Targets

- ☐ I can determine the sample space for a repeated experiment using lists, tables, and tree diagrams.
- ☐ I can determine the sample space for a two-step experiment.

14.2.1 Warm-Up: Bell Work

1. The integer chips shown each have a yellow side to represent a positive integer and a red side to represent a negative integer.



Complete the table for the random process below.

Number of Flips of an Integer Chip	Sample Space
1	
2	
3	

14.2.2 Collaboration: Outcome Tables

Another way to organize the outcomes in a sample space is with an outcome table.



Outcome tables are a useful way to list and organize all possible outcomes in a random process with two steps. They are especially useful when there are a large number of outcomes in the sample space.

One limitation of an outcome table is that it cannot be used to determine the sample space of experiments with more than two steps.

Below is an outcome table showing the possible outcomes when a coin is flipped two times.

		Toss 2	
		Heads	Tails
Toss 1	Heads	HH	HT
	Tails	TH	TT

Since there are six possible outcomes when one standard six-sided die is rolled, the outcome table for rolling two six-sided dice needs to be six by six.

In the outcome table shown, the numbers 1-6 in the second column from the left represent the possible outcomes when the first die is rolled.

- 1. Complete the grey row to represent the possible outcomes from rolling the second die.

		Die 2					
		1		3			6
Die 1	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2		(2, 2)		(2, 4)	(2, 5)	
	3						
	4						
	5						
	6						

The pairs of numbers in the first white row of the outcome table represent all outcomes that can occur when a 1 is rolled on the first die. The outcome (1, 3) means the first die rolled is a 1, and the second die rolled is a 3.

2. Complete the next row of the outcome table to show the possible outcomes when a 2 is rolled on the first die.
3. What patterns do you see in the outcome table at this time?
4. Work with your partner to complete the rest of the outcome table.
5. When your outcome table is complete, find another pair of students to check your work with. Make any necessary changes.
6. Complete the table.

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Rolling a fair die two times			

7. How many outcomes are in Event A = rolling the same number?
8. Work with your partner to complete the table showing the strengths and weaknesses/limitations of tree diagrams and outcome tables.

	Outcome Table	Tree Diagram
Strengths		
Weaknesses/ Limitations		

14.2.3 Guided Practice: Determining the Number of Outcomes in the Sample Space

1. A fair coin is tossed, and a fair six-sided die is rolled. Determine the sample space using either a tree diagram or an outcome table.

2. Complete the table.

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Tossing a fair coin and rolling a fair die			

3. Work with your partner to summarize the relationship between the number of outcomes in each step of a repeated experiment and the total number of outcomes in the experiment.

14.2.4 Bring It Together: Determining the Sample Space

1. Fill in the blue boxes with the same operation symbol to make all of the number sentences true.

Experiments	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Flip two fair coins	2	2	= 4
Roll two fair six-sided dice	6	6	= 36
Flip a fair coin and roll a fair die	2	6	= 12

2. To determine the sample space for a multi-step experiment, it can be helpful to know how many total outcomes should be in the sample space. To figure this out,

- add
 - multiply

 the number of outcomes in each step.
3. For example, if an experiment consists of flipping a fair coin and spinning a spinner with eight different outcomes, the total number of outcomes in the sample space can be found by

- multiplying
 - adding

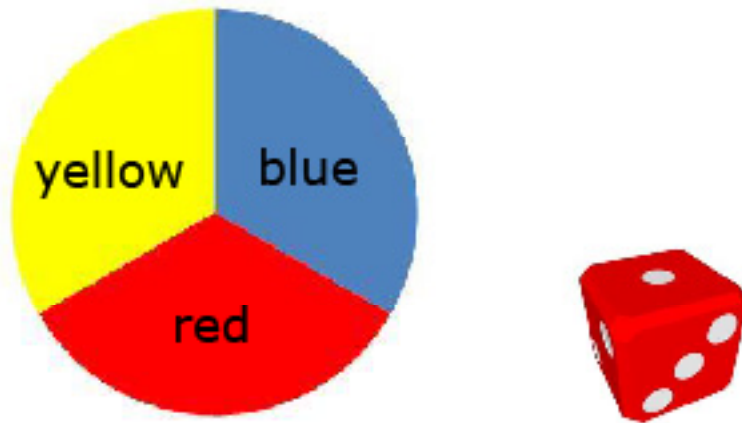
 the number of outcomes in each step of the repeated experiment. In this case,

- $2 \cdot 8 = 16$.
 - $2 + 8 = 10$.
4. If a fair coin is tossed six times, the total number of outcomes in the sample space of this repeated experiment is

- $6 \cdot 2 = 12$.
 - $2^6 = 64$.

14.2.5 Your Turn!

Marianne and three friends are playing a game that requires them to spin the spinner shown, then roll a fair six-sided die on each person's turn to play.

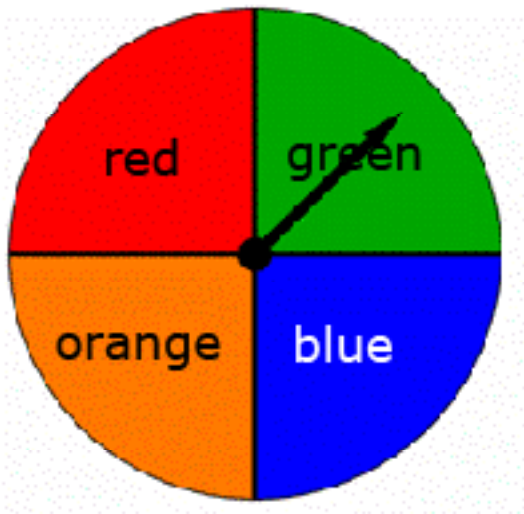


1. How many outcomes are in the sample space when...
 - a. ...the spinner is spun once?
 - b. ...a fair six-sided die is rolled once?
 - c. ...the spinner is spun, then a six-sided die is rolled?
2. Use a tree diagram or outcome table to determine the sample space.
3. Write out the sample space for one player's turn.

14.2.6 Wrap-Up: Ticket Out the Door

Spinner A is shown.

Spinner A



1. Determine the number of outcomes in the sample space when **Spinner A** is spun twice.
2. Explain which method you would choose to determine the sample space for this random process: a list, table, or tree diagram.
3. Write the sample space for this repeated experiment.